

Monocular depth estimation

Project Option 3

Monocular depth estimation is the process of predicting the distance of objects from the camera using only a single RGB image. Unlike stereo vision, which relies on multiple viewpoints to triangulate depth, monocular depth estimation leverages visual cues such as texture gradients, object sizes, occlusions, and perspective to infer the scene's three-dimensional structure.

Why Tackle Monocular Depth Estimation?

- **Widespread Applications:**
 - **Autonomous Driving:** Vehicles can better understand their surroundings, identify obstacles, and plan safer navigation strategies using accurate depth maps.
 - **Robotics:** Robots can navigate and interact within complex environments without requiring bulky sensor arrays, leading to more compact and cost-effective solutions.
 - **Augmented Reality:** AR systems benefit from depth information to seamlessly blend virtual objects with real-world scenes, enhancing user experiences with more realistic interactions.
 - **Scene Understanding:** Depth estimation supports advanced image segmentation, object detection, and scene reconstruction, contributing to a wide range of computer vision applications.
- **Cost and Hardware Efficiency:**
 - Single-camera systems are cheaper and simpler to deploy compared to multi-camera setups or depth sensors, making this technology accessible for consumer devices and mass-market applications.

By solving the problem of monocular depth estimation, researchers and practitioners can unlock more intuitive and efficient ways to interpret visual data, leading to innovations across industries and contributing to the advancement of intelligent systems.

Competition

To join the Kaggle competition, please use the private links shared on the Moodle page.

Training Data

In this project, you are tasked with predicting depth maps from everyday scenes captured in RGB images. We provide a robust training dataset consisting of 22,605 samples of size 560x560. Each training sample includes an input RGB image and its corresponding target depth map. The depth maps are stored as NumPy arrays containing pixel-wise depth measurements in meters, with values ranging from a minimum of 1 millimeter up to a

maximum of 80 meters. But depth scales between images are not meaningful and only represent ordering within individual images

Your model's performance will be evaluated using a separate test set of 591 samples of size 560x560. Note that the ground truth depth maps for these test samples are not provided, ensuring that your model is truly tested on its ability to generalize.

After generating your predictions, save each depth map as a NumPy array (.npy) file with the appropriate filename. Then, run the provided submission generation code - pointing it to the folder containing your prediction files - to create a CSV file for submission to the Kaggle competition.

The data can be accessed on Kaggle

Evaluation Metrics

Your approach is evaluated according to scale-invariant Root Mean Squared Error (RMSE). See a detailed description on Kaggle.